



Name: _____

SCEGGS Darlinghurst

Preliminary Year, 2003
Semester 2 Examination

Mathematics (Extension)

General Instructions

- Reading time – 5 minutes
- Working time – 1½ hours
- This paper has **5 questions**
- Attempt **all** questions
- Answer all questions on the pad paper provided
- Write your name on every page
- Total marks for all parts (62)
- Approved calculators may be used

Questions 1-5

Total marks (62)

- Attempt all parts of Questions 1-5

	Communication	Calculus	Reasoning	Total
Question 1	/3		/2	/12
Question 2		/1	/5	/12
Question 3	/3	/2		/13
Question 4			/9	/12
Question 5	/3		/10	/13
TOTAL	/9	/3	/26	/62

- Answer the questions on the pad paper provided
- Clearly label each part
- Write your name on the top of each page
- START EACH QUESTION ON A NEW PAGE

Question 1 (12 Marks)

Marks

(a) Solve for x :

3

$$\frac{3x+1}{2-x} \geq 1$$

(b) A and B are the points $(-1, 3)$ and $(2, 0)$ respectively:

(i) Find the co-ordinates of the point P which divides AB externally in the ratio 5:2.

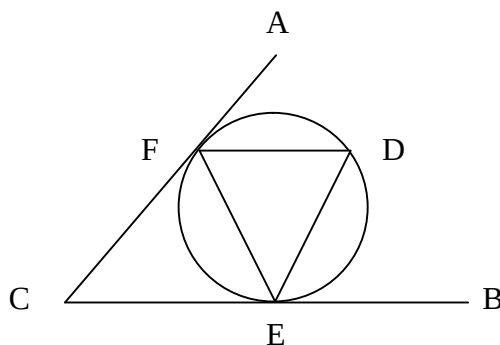
2

(ii) If the line AB cuts the y axis at Q, find the ratio in which Q divides AB internally without finding the equations of any lines.

2

(c) In the diagram, AC and BC are tangents to the circle, touching the circle at F and E respectively. $\angle ACB$ equals 50° . Copy the diagram onto your paper.

Show that $\angle CEF$ is 65° and hence find $\angle EDF$.



3

(d) Prove the identity $\frac{1 - \tan^2 A}{1 + \tan^2 A} = \cos 2A$.

2

-
- START A NEW PAGE
-

Question 2 (12 Marks)**Marks**

(a) The polynomial $P(x) = x^3 + x^2 + x - 2$ has roots, α, β, γ .

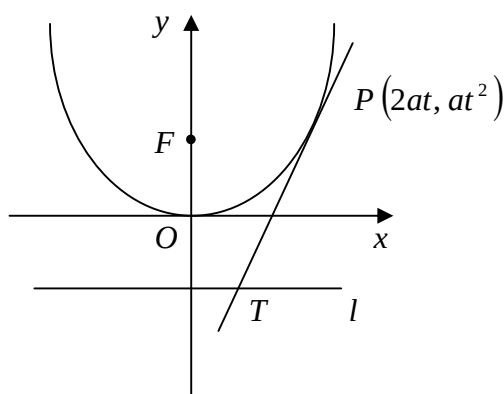
(i) Find the value of $\alpha\beta\gamma$. **1**

(ii) If $\alpha = 1$, find the value of $\frac{1}{\beta} + \frac{1}{\gamma}$. **2**

(b) (i) Find the gradient of the tangent to the curve $y = x^2 + 3$ at the point $(1, 4)$. **1**

(ii) Find the acute angle between the line $y = 3x + 1$ and the curve $y = x^2 + 3$ at the point of intersection $(1, 4)$. Give your answer to the nearest minute. **2**

(c) The tangent $tx - y - at^2 = 0$ at the point $P(2at, at^2)$ on the parabola $x^2 = 4ay$ cuts the directrix l at T . F is the focus of the parabola.



(i) Find the co-ordinates of T . **1**

(ii) Show that TF is perpendicular to PF . **3**

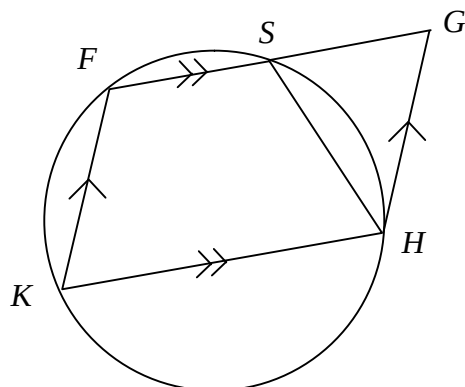
Question 2 continues on the next page

Question 2 (continued)

Marks

(d)

2



The figure $FGHK$ is a parallelogram. The point S lies on FG , and F, S, H, K lie on a circle.

Copy or trace the diagram onto your answer paper.

Prove that triangle HGS is isosceles.

-
- START A NEW PAGE
-

Question 3 (13 Marks)

Marks

(a) Consider the polynomial $P(x) = 6x^3 - 5x^2 - 2x + 1$.

(i) Show that 1 is a zero of $P(x)$.

1

(ii) Express $P(x)$ as a product of 3 linear factors.

2

(iii) Solve the inequality $P(x) \leq 0$.

1

(b) Using the substitution $\frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{1}{\sqrt{x+h} + \sqrt{x}}$ find from first principles the derivative of $y = \sqrt{x}$.

2

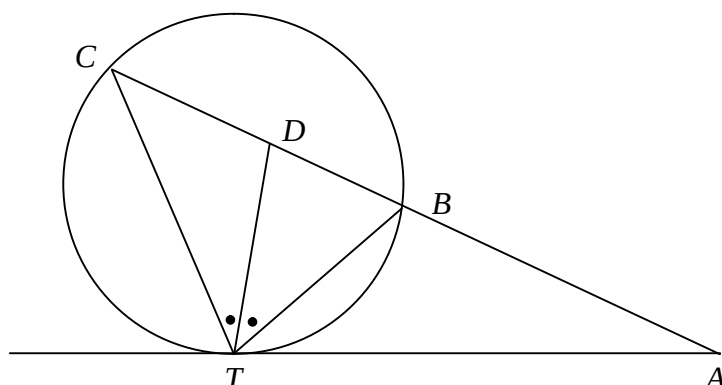
(c) (i) Express $4\sin\theta - 3\cos\theta$ in the form $A\sin(\theta - \alpha)$ where $A > 0$ and $0^\circ < \alpha < 90^\circ$. Give α to the nearest degree.

2

(ii) Find all solutions of $4\sin\theta - 3\cos\theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$. Give θ to the nearest degree.

2

(d)



TA is a tangent to a circle. Line $ABDC$ intersects the circle at B and C . Line TD bisects angle BTC .

3

Prove $AT = AD$.

-
- START A NEW PAGE
-

Question 4 (12 Marks)

Marks

- (a) ✓ (i) Using the $\sin(A - B)$ formula, with $A = 60^\circ$ and $B = 45^\circ$ (or otherwise), find the value of $\sin 15^\circ$ in surd form.

2

- ✓ (ii) Prove the identity:

2

$$\tan \alpha = \operatorname{cosec} 2\alpha - \cot 2\alpha$$

- ✓ (iii) Using the results in part (ii), find a value (in surd form) for $\tan 15^\circ$.

2

- ✓ (iv) Using the results in parts (i), (ii) and (iii), (or otherwise) show that:.

3

$$\tan 7\frac{1}{2}^\circ = \sqrt{6} + \sqrt{2} - \sqrt{3} - 2$$

- (b) The remainder when $x^3 + ax + b$ is divided by $(x - 2)(x + 3)$ is $2x + 1$. Find the values of a and b .

3

-
- START A NEW PAGE
-

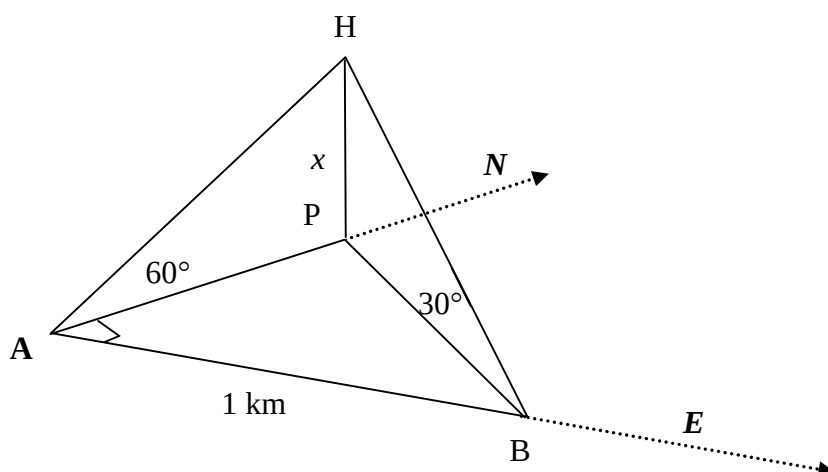
Question 5 (13 Marks)

Marks

- (a) On the same set of axes, draw the graphs of $y = |x - 2|$ and $y = x + 1$.
For what values of x is $|x - 2| < x + 1$?

3

- (b) Anna (A) is standing due south of Phillip (P) who is assisting an injured bushwalker. A rescue helicopter (H) is hovering directly over P and lowering a stretcher. Anna measures the angle of elevation of the helicopter to be 60° from her position. Belinda (B) is 1 kilometre due east of A and measures the angle of elevation of the helicopter to be 30° . The height of the helicopter above P is x metres.



- (i) Write expressions for both AP and BP in terms of x .
- (ii) Hence or otherwise, find the height of the helicopter (x), correct to the nearest m.
- (c) $P(2ap, ap^2)$ is a variable point on a parabola $x^2 = 4ay$. The line PR is perpendicular to the directrix of the parabola with R on the directrix. The tangent to the parabola at P meets the y -axis at T . If M is the midpoint of the interval RT :
- (i) find the locus of M .
- (ii) show that the directrix of the locus of M is the x -axis.

1

3

4

2

End of Paper

① a) $\frac{3x+1}{2-x} \geq 1$

undefined when $x=2$

multiply both sides by $(2-x)^2$

$$\frac{3x+1}{2-x} \times (2-x)^2 \geq 1 \times (2-x)^2$$

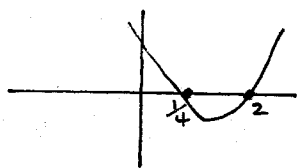
$$(3x+1)(2-x) \geq (2-x)^2$$

$$6x - 3x^2 + 2 - x \geq 4 - 4x + x^2$$

$$0 \geq 4x^2 - 9x + 2$$

$$4x^2 - 9x + 2 \leq 0$$

$$(4x-1)(x-2) \leq 0$$



$$\frac{1}{4} \leq x \leq 2$$

b) External division 5:-2

A(-1,3) B(2,0)
5:-2

$$x = \frac{-2x-1+5x2}{5-2}$$

$$y = \frac{-2x3+5x0}{5-2}$$

$$= \frac{2+10}{3}$$

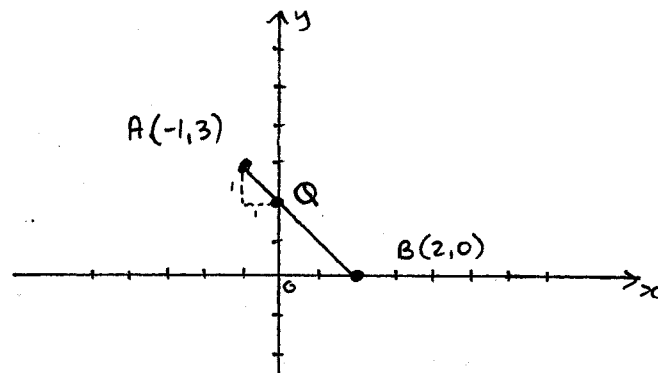
$$= \frac{-6+0}{3}$$

$$= \frac{12}{3}$$

$$= -2$$

$$= 4$$

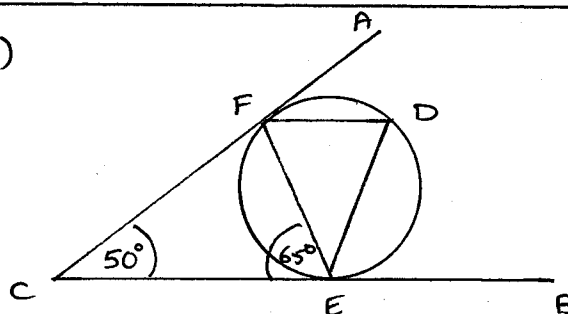
$$\therefore P(4, -2)$$



By observation

Q divides AB internally in the ratio 1:2

c)



$$CF = CE$$

Since tangents from an exterior point are equal.

$\therefore \triangle CFE$ is isosceles.

$$\therefore \angle CEF = \frac{1}{2}(180^\circ - 50^\circ) \quad (\text{sum } \Delta = 180^\circ)$$

$$= \frac{1}{2} \times 130^\circ$$

$$= 65^\circ$$

$$\angle EDF = \angle CEF$$

$$= 65^\circ$$

(Angles in the alternate segments are equal.)

Com 3

(2)

$$d) \text{ LHS} = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$= \frac{1 - \frac{\sin^2 A}{\cos^2 A}}{1 + \frac{\sin^2 A}{\cos^2 A}}$$

$$= \frac{\frac{\cos^2 A - \sin^2 A}{\cos^2 A}}{\frac{\cos^2 A + \sin^2 A}{\cos^2 A}}$$

$$\frac{\cos^2 A - \sin^2 A}{\cos^2 A}$$

$$= \frac{\frac{\cos^2 A - \sin^2 A}{\cos^2 A}}{\frac{1}{\cos^2 A}}$$

$$= \frac{\cos^2 A - \sin^2 A}{\cos^2 A} \times \frac{\cos^2 A}{1}$$

$$= \cos^2 A - \sin^2 A$$

$$= \cos 2A$$

Reas 2

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$\text{put } \alpha = 1$$

$$1 + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{1}{2}$$

$$\frac{1}{\beta} + \frac{1}{\gamma} = -\frac{1}{2}$$

$$b) \text{ i) } y = x^2 + 3$$

$$y = 2x$$

At (1,4) gradient tangent

$$m_1 = 2$$

Calculus 1

$$\text{ii) } y = 3x + 1$$

$$m_2 = 3$$

Acute angle

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{2 - 3}{1 + 2 \times 3} \right|$$

$$= \left| \frac{-1}{7} \right|$$

$$= \frac{1}{7}$$

$$\theta \doteq 8^\circ 8' (\text{to nearest minute})$$

$$\textcircled{2} P(x) = x^3 + x^2 + x - 2$$

Roots α, β, γ i) Product of roots

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$= -\frac{-2}{1}$$

$$= 2$$

ii) Sum of Roots

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

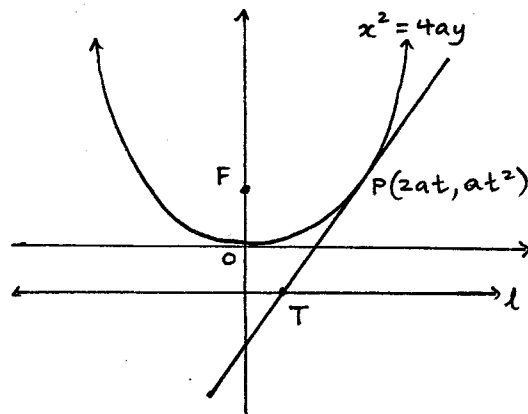
$$= -1$$

Sum two at a time

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$= 1$$

c)



Tangent at $P(2at, at^2)$

$$tx - y - at^2 = 0$$

cuts the directrix at T

directrix is $y = -a$

Solve simultaneously

$$\begin{aligned} tx - y - at^2 &= 0 & (1) \\ y &= -a & (2) \end{aligned}$$

Substitute (2) into (1)

$$tx + a - at^2 = 0$$

$$tx = at^2 - a$$

$$x = \frac{a(t^2 - 1)}{t}$$

T has coordinates

$$\left(\frac{a(t^2 - 1)}{t}, -a \right)$$

ii) $\left[\begin{array}{l} \text{Focus } F(0, a) \\ P(2at, at^2) \end{array} \right]$

Gradient TF

$$m_1 = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-a - a}{\frac{a(t^2 - 1)}{t} - 0}$$

$$= \frac{-2a}{\frac{a(t^2 - 1)}{t}}$$

$$= \frac{-2t}{t^2 - 1}$$

Gradient PF

$$m_2 = \frac{at^2 - a}{2at - 0}$$

$$= \frac{a(t^2 - 1)}{2at}$$

$$= \frac{t^2 - 1}{2t}$$

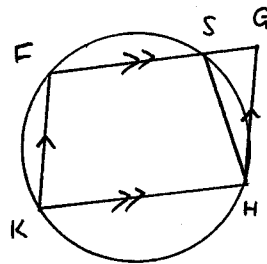
$$\text{Since } m_1 \times m_2 = \frac{-2t}{t^2 - 1} \times \frac{t^2 - 1}{2t}$$

$$= -1$$

$$\therefore TF \perp PF$$

Ques 3

d)



FGHK is a parallelogram

$$\therefore \angle FKH = \angle FGH \quad (\text{opposite } \angle \text{s in a parm.})$$

$$= x$$

F, S, H, K lie on a circle

$\therefore$ FSHK is a cyclic quadrilateral

$$\therefore \angle FKH + \angle FSH = 180^\circ$$

(opp. $\angle$ s in a cyclic quad. are supplementary)

$$\begin{aligned} \angle FSH &= 180^\circ - \angle FKH \\ &= 180^\circ - x \end{aligned}$$

$$\begin{aligned} \angle GSH &= 180^\circ - \angle FSH & (\angle \text{sum straight } \angle = 180^\circ) \\ &= 180^\circ - (180^\circ - x) \\ &= x \end{aligned}$$

$$\therefore \angle SGH = \angle GSH = x$$

Ques 2

$\therefore \triangle HGS$ is isosceles since it has 2 equal angles.

Q3

a) $P(x) = 6x^3 - 5x^2 - 2x + 1$

i) $P(1) = 6 - 5 - 2 + 1$
 $= 0$

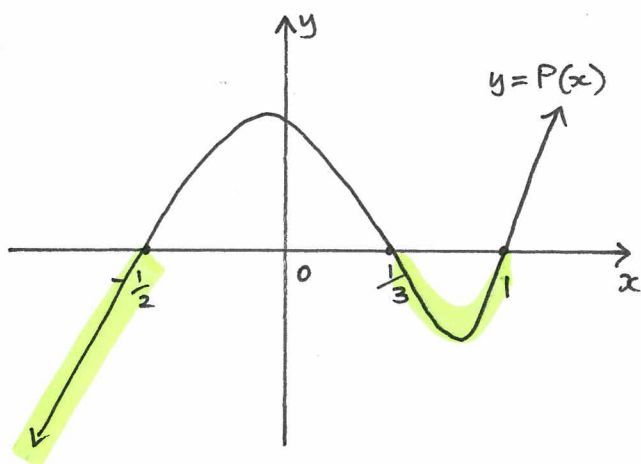
$\therefore x=1$ is a zero of $P(x)$

$\therefore (x-1)$ is a factor of $P(x)$

ii)
$$\begin{array}{r} 6x^2 + x - 1 \\ x-1 \overline{) 6x^3 - 5x^2 - 2x + 1} \\ \underline{6x^3 - 6x^2} \\ x^2 - x + 1 \\ \underline{x^2 - x} \\ -x + 1 \\ \underline{-x + 1} \\ 0 \end{array}$$

$\therefore P(x) = (x-1)(6x^2 + x - 1)$
 $= (x-1)(3x-1)(2x+1)$

iii)



$P(x) \leq 0$
 for

$x \leq -\frac{1}{2}, \frac{1}{3} \leq x \leq 1$

b) $y = \sqrt{x}$

$f(x) = \sqrt{x}$

$f(x+h) = \sqrt{x+h}$

Using first principles

$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$

using the given substitution

$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$

$= \frac{1}{\sqrt{x+0} + \sqrt{x}}$

$= \frac{1}{2\sqrt{x}}$

Calc
2

c) i) $4 \sin \theta - 3 \cos \theta$

$= A \sin(\theta - \alpha)$

$= A \sin \theta \cos \alpha - A \cos \theta \sin \alpha$

Match parts

$A \cos \alpha = 4$ ①

$A \sin \alpha = 3$ ②

Find A

$A^2 = 4^2 + 3^2$

$A = \sqrt{16 + 9}$
 $= \sqrt{25}$
 $= 5$

Find α

divide ②/①

$\tan \alpha = \frac{3}{4}$

$\therefore \alpha \approx 37^\circ$

(5)

$$\therefore 4 \sin \theta - 3 \cos \theta$$

$$= 5 \sin(\theta - 37^\circ)$$

$$\text{ii) } 4 \sin \theta - 3 \cos \theta = 1$$

$$5 \sin(\theta - 37^\circ) = 1$$

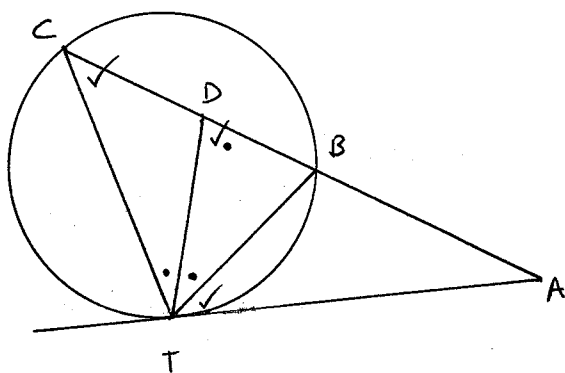
$$\sin(\theta - 37^\circ) = \frac{1}{5}$$

$$\therefore \theta - 37^\circ = 11^\circ 32' \text{ or } 168^\circ 28'$$

$$\therefore \theta = 48^\circ \text{ or } 205^\circ$$

✓✓

d)



$$\angle CTD = \angle BTD \quad (\text{given})$$

$$\angle ATB = \angle TCD \quad (\angle \text{ in the alt segment} = \angle \text{ between chord + tangent})$$

$$\angle TDB = \angle CTD + \angle TCD \quad (\text{ext. } \angle \text{ in } \triangle CTD)$$

$$\therefore \angle ATD = \angle ADT$$

$$\therefore AT = AD \quad (\text{sides opp. } \angle \text{ in an isosceles } \triangle =)$$

QED
3

$$\textcircled{4} \text{ a) i) } \sin(A-B)$$

$$= \sin A \cos B - \cos A \sin B$$

$$\sin 15^\circ = \sin(45^\circ - 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2} \quad \checkmark$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} \quad \checkmark$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$\text{ii) Prove } \tan \alpha = \operatorname{cosec} 2\alpha - \cot 2\alpha$$

$$\text{RHS} = \operatorname{cosec} 2\alpha - \cot 2\alpha$$

$$= \frac{1}{\sin 2\alpha} - \frac{\cos 2\alpha}{\sin 2\alpha}$$

$$= \frac{1 - \cos 2\alpha}{\sin 2\alpha}$$

$$= \frac{1 - (\cos^2 \alpha - \sin^2 \alpha)}{2 \sin \alpha \cos \alpha} \quad \checkmark$$

$$= \frac{1 - \cos^2 \alpha + \sin^2 \alpha}{2 \sin \alpha \cos \alpha}$$

$$\text{Substituting } 1 - \cos^2 \alpha = \sin^2 \alpha$$

$$= \frac{\sin^2 \alpha + \sin^2 \alpha}{2 \sin \alpha \cos \alpha}$$

$$= \frac{2 \sin^2 \alpha}{2 \sin \alpha \cos \alpha}$$

$$= \frac{\sin \alpha}{\cos \alpha} \quad \checkmark$$

$$= \tan \alpha$$

$$= \text{LHS}$$

$$\text{iii) } \tan 15^\circ = \operatorname{cosec} 30^\circ - \cot 30^\circ$$

$$= \frac{1}{\sin 30^\circ} - \frac{1}{\tan 30^\circ} \quad \checkmark$$

$$= \frac{1}{\frac{1}{2}} - \frac{1}{\frac{1}{\sqrt{3}}} \quad \checkmark$$

$$= 2 - \sqrt{3} \quad \checkmark$$

$$\text{iv) } \tan 7\frac{1}{2}^\circ = \operatorname{cosec} 15^\circ - \cot 15^\circ$$

$$= \frac{1}{\sin 15^\circ} - \frac{1}{\tan 15^\circ}$$

$$= \frac{1}{\frac{\sqrt{6}-\sqrt{2}}{4}} - \frac{1}{2-\sqrt{3}} \quad \checkmark$$

$$= \frac{4}{\sqrt{6}-\sqrt{2}} - \frac{1}{2-\sqrt{3}}$$

Rationalise each denominator

$$\frac{4}{\sqrt{6}-\sqrt{2}} \times \frac{\sqrt{6}+\sqrt{2}}{\sqrt{6}+\sqrt{2}} = \frac{4(\sqrt{6}+\sqrt{2})}{6-2}$$

$$= \sqrt{6}+\sqrt{2}$$

$$\frac{1}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{2+\sqrt{3}}{4-3}$$

$$= 2+\sqrt{3} \quad \checkmark$$

$$\therefore = \sqrt{6}+\sqrt{2} - (2+\sqrt{3})$$

$$= \sqrt{6}+\sqrt{2} - \sqrt{3} - 2 \quad \checkmark$$

Reas 9

$$\text{b) } P(x) = A(x) \cdot Q(x) + R(x)$$

$$x^3 + ax + b = (x-2)(x+3) \cdot Q(x) + 2x+1$$

Substitute $x=2$

$$8+a+b = 0 + 4+1$$

$$a+b = -3 \quad \textcircled{1}$$

Substitute $x=-3$

$$-27 - 3a + b = 0 - 6 + 1$$

$$-3a + b = 22 \quad \textcircled{2} \quad \checkmark$$

Solve simultaneously

$$a+b = -3 \quad \textcircled{1}$$

$$-3a+b = 22 \quad \textcircled{2}$$

Subtract

$$4a = -25$$

$$a = -25/4 \quad \checkmark$$

$$a = -6\frac{1}{4}$$

subst. into ①

$$-6\frac{1}{4} + b = -3$$

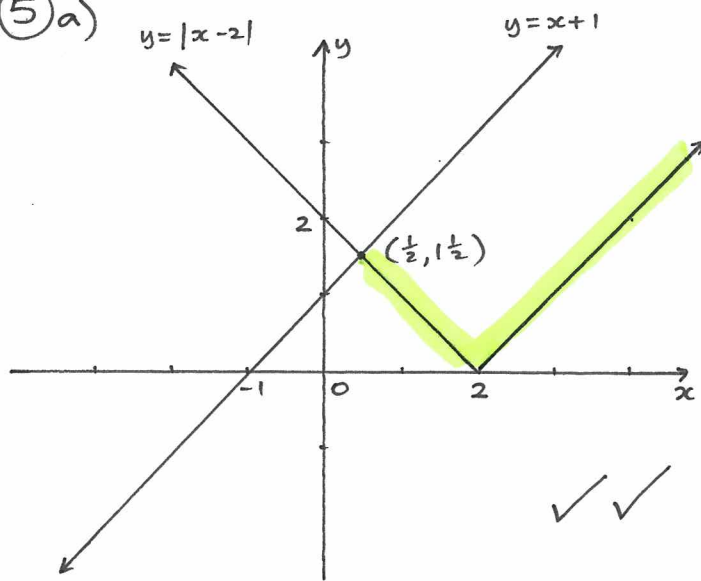
$$b = 3\frac{1}{4} \quad \checkmark$$

Solution

$$\therefore a = -6\frac{1}{4}$$

$$b = 3\frac{1}{4}$$

⑤ a)



One point of intersection only

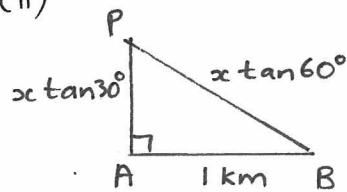
when $x+1 = -(x-2)$
 $x+1 = -x+2$
 $2x = 1$
 $x = 1/2$

$(\frac{1}{2}, 1\frac{1}{2})$

from the graph
 $|x-2| < x+1$
 for $x > \frac{1}{2}$

$\frac{6m}{3}$

(ii)



Using Pythagoras rule

$$BP^2 = AP^2 + AB^2$$

$$x^2 \tan^2 60^\circ = x^2 \tan^2 30^\circ + 1^2 \quad \checkmark$$

$$x^2 \tan^2 60^\circ - x^2 \tan^2 30^\circ = 1$$

$$x^2 (\tan^2 60^\circ - \tan^2 30^\circ) = 1$$

$$x^2 = \frac{1}{\tan^2 60^\circ - \tan^2 30^\circ}$$

$$x = \sqrt{\frac{1}{\tan^2 60^\circ - \tan^2 30^\circ}} \quad \checkmark$$

$$= \sqrt{\frac{1}{3 - \frac{1}{3}}}$$

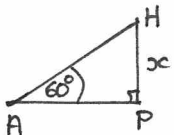
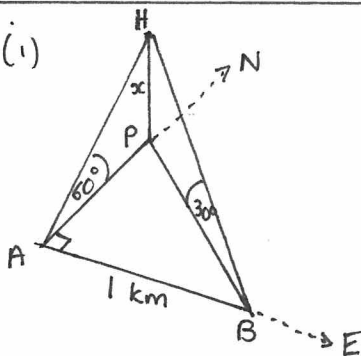
$$= \sqrt{\frac{3}{8}}$$

$$\doteq 0.612372 \dots \text{ km}$$

$$\doteq 612 \text{ m} \quad \checkmark$$

(to nearest metre)

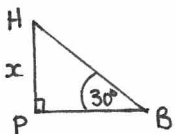
b) (i)



$$\tan 60^\circ = \frac{x}{AP}$$

$$AP = \frac{x}{\tan 60^\circ}$$

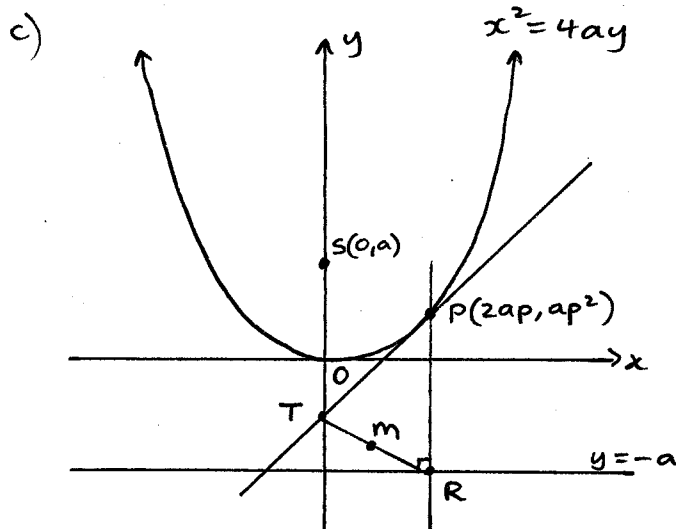
or $AP = x \tan 30^\circ$



$$\tan 30^\circ = \frac{x}{BP}$$

$$BP = \frac{x}{\tan 30^\circ}$$

or $BP = x \tan 60^\circ$



$$x^2 = 4ay$$

directrix $y = -a$

Coordinates of R
(2ap, -a)

Equation of tangent at P

$$y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

At P(2ap, ap^2)

tangent $m_1 = \frac{2ap}{2a}$

$$= p$$

$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$

Coordinates of T

Tangent cuts y-axis when $x = 0$

$$y = px - ap^2$$

$$y = 0 - ap^2$$

$$y = -ap^2$$

$$\therefore T(0, -ap^2)$$

Midpoint of TR

$$T(0, -ap^2) \quad R(2ap, -a)$$

$$m = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{0 + 2ap}{2}, \frac{-ap^2 - a}{2} \right)$$

$$= \left(ap, -\frac{a(p^2 + 1)}{2} \right)$$

Locus of m

$$\begin{cases} x = ap \\ y = -\frac{a}{2}(p^2 + 1) \end{cases}$$

Substitute $p = \frac{x}{a}$ into y

$$y = -\frac{a}{2} \left(\left(\frac{x}{a} \right)^2 + 1 \right)$$

$$y = -\frac{a}{2} \left(\frac{x^2}{a^2} + 1 \right)$$

$$y = -\frac{x^2}{2a} - \frac{a}{2}$$

The locus is a parabola.

$$(x^2) \quad 2ay = -x^2 - a^2$$

$$x^2 = -2ay - a^2$$

$$x^2 = -2a \left(y + \frac{1}{2}a \right)$$

$$x^2 = -4 \cdot \frac{1}{2}a \left(y + \frac{1}{2}a \right)$$

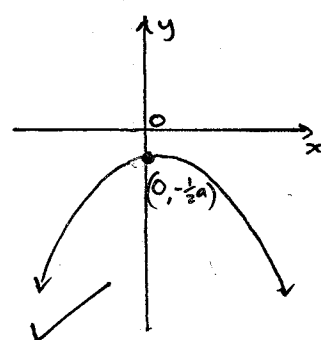
The locus is a parabola.

Vertex $(0, -\frac{1}{2}a)$

Focal length $= \frac{1}{2}a$

Directrix $y = 0$

$\therefore$ The directrix is the x-axis.



Ans
10